Md Sabbir Ahmed Khan

Senior Software Engineer | **Kotlin** | **Java** | **Android** | **Kotlin Multiplatform**
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SUMMARY

Senior Software Engineer with 12+ years of building and shipping software for fintech, health and enterprise products. Own the full app lifecycle, from idea and system design to testing and Play Store release. Currently leading the Android work at a Singapore-based drone company, rebuilding the main app from scratch with multi-module MVVM architecture, Coroutines and Jetpack Compose, supported by AI-assisted development. Previously shipped apps for fintech, health-tech (Samsung), and hardware-connected products like IoT devices.

SKILLS

Programming Languages: Kotlin, Java, Python, Swift, Objective C
Android: Jetpack Compose, Coroutines & Flow, Hilt/Dagger, Room, Retrofit, WorkManager, Material Design
Cross-platform: Kotlin Multiplatform, Compose Multiplatform, Wear OS
Architecture: Multi-module, MVVM, MVI, Clean Architecture, Offline-first, Baseline Profiles
Security: Android Keystore, AES, BiometricPrompt, OAuth 2.0, Certificate Pinning, R8/ProGuard
Testing & Delivery: JUnit, MockK, Turbine, Espresso, Compose UI Testing, CI/CD, Fastlane, Gradle
Methodologies & Tools: Agile/Scrum, Sprint Planning, Jira, Figma, Code Review, Mentoring
Integrations: Firebase, Payment Gateways, REST APIs, TCP/UDP Sockets, BLE, Android NDK
AI Tools: Claude Code, Codex, Cursor, GitHub Copilot, OpenAI API, Gemini API

EXPERIENCE

Senior Android Engineer | Avetics Global Pte Ltd | Dec 2022 - Present | Singapore
Singapore tech company building enterprise apps for inspection and monitoring.

- Rebuilt the official Android app from scratch in Kotlin, Jetpack Compose, Coroutines & Flow and Hilt, with DJI Mobile SDK for live camera feeds, autonomous flight and low-latency remote drone control.
- Designed the app's multi-module MVVM/MVI architecture and it is now the standard used company-wide for multiple drone projects.
- Lead the Android work hands-on: build core features, plan sprints in Jira (Agile/Scrum), turn Figma designs into app screens and review code.
- Brought AI-assisted development (Claude Code, Gemini CLI, Cursor) to the team by streamlining the work process and setting up a Standard Operating Procedure (SOP) for development.
- Built a secure firmware update flow for in-house IoT hardware: firmware is downloaded from the server and verified with an AES key protected by Android Keystore.
- Set up testing for the app: JUnit, MockK and Turbine for ViewModel and Flow logic, Hilt test modules that swap the DJI SDK and socket layers with fakes, and Compose UI tests.
- Used Firebase Crashlytics and LeakCanary to monitor and fix crashes and memory leaks, cutting production crashes by about 90%.
- Designed TCP socket infrastructure for real-time, low-latency hardware communication, streaming live data to the UI with Kotlin Flow, plus an offline-first Room data layer, Firebase Auth and Realtime Database.

Senior Android Engineer | GrowFolio (formerly Plentina) | Feb 2022 - Nov 2022 | Philippines
Silicon Valley-registered fintech with a BNPL app for the Philippines and Vietnam. Worked remotely from Bangladesh.

- Developed a Buy Now, Pay Later (BNPL) Android app across two Southeast Asian markets using Kotlin, Jetpack Compose, and Coroutines.
- Built Kotlin Multiplatform (KMM) infrastructure and streamlined CI/CD with Jenkins and Fastlane to automate builds, testing, and Play Store/App Store releases.
- Built the KYC onboarding flow for identity verification of new users.
- Integrated several third-party payment gateways into the BNPL checkout flow, handling payment results and errors so users in the Philippines and Vietnam could complete purchases smoothly.
- Migrated QR code scanning from the deprecated Google Mobile Vision API to Google ML Kit's on-device barcode scanning to improve scan accuracy.
- Optimized asynchronous state management with Kotlin Flow and implemented unit, integration, and UI tests using MockK and Turbine.

Senior Software Engineer II | Samsung R&D Institute Bangladesh | Oct 2019 - Feb 2022 | Bangladesh

- Developed Android health-tracking apps for Samsung's health ecosystem with server-side sync and Room persistence.
- Built Wear OS companion apps for health activity monitoring and Bluetooth Low Energy (BLE) based Tizen watch app upgrades with seamless connectivity and resumable transfers.
- Implemented timezone-change broadcast receivers for automatic watch timezone synchronization and Google Cloud Messaging (GCM) for push notifications.
- Wrote JUnit, Mockito, Espresso, and in-memory Room tests, using fake sensor data to validate Wear OS features without physical devices.
- Contributed to Samsung SmartThings iOS app development using Swift.

Android Application Developer | Kutumbita Bangladesh Ltd | Nov 2018 - Sep 2019 | Bangladesh

- Built Kinship.ai, the company's internal chat app, as the sole Android developer, implementing real-time messaging with Socket.IO and chatbot server integration.
- Owned the full app lifecycle, from design and development to testing and Google Play release, using Java and Kotlin while migrating legacy code to Kotlin with MVVM and LiveData.
- Implemented asynchronous Room database backups to the server using WorkManager.
- Tested socket reconnection with a fake socket client and wrote unit and UI tests using JUnit, Mockito, and Espresso.

Founder and Mobile Engineer | Power GP BD Limited | Jan 2012 - Oct 2018 | Bangladesh

- Founded and led a mobile company, delivering 100+ Android apps in Java (Eclipse IDE) for phones, tablets and Android TV, adapting UI to each screen size for enterprise, B2B and consumer clients.
- Partnered with clients and stakeholders to translate ideas into applications, owning the full lifecycle from requirements analysis and system design to development and deployment.
- Received two Facebook Startup Program awards for high-impact mobile application development.

SELECTED KEY PROJECTS

Volarious - Enterprise Drone Control App | Dec 2022 - Present

- Built shared Android modules that are reused across the DJI, Vesper and Autel drone apps.
- Wrapped the Java-based DJI Mobile SDK in Kotlin Coroutines and Flow for clean, testable calls.
- Ran a C++ processing library through the Android NDK inside the Kotlin app.
- Built live camera view, waypoint missions and GPS-based autonomous flight on tablet screens.

V-Assist - Line Status Monitoring App | Apr 2024 - Present

- Built in Jetpack Compose with MVVM and published on Google Play.
- Built TCP and UDP messaging for fast, two-way communication between the app and the hardware.
- Added Baseline Profiles, cutting app launch time and UI latency.
- Used AI-assisted development (Claude Code, Gemini CLI) for design, coding, debugging and tests.

Samsung Health with Wear OS | Oct 2019 - Jan 2022

- Built a Wear OS companion that syncs health activity data between the watch and phone over BLE.
- Used RxJava to sync data between the watch, local Room storage and the server in the background.
- Tested Wear OS features with fake sensor data, without needing a physical watch.

Plentina - Buy Now, Pay Later | Feb 2022 - Nov 2022

- Early adopter of Kotlin Multiplatform (KMM): built the iOS version of the same app with shared ViewModels.
- Converted the existing XML-based UI to Jetpack Compose for a future-proof codebase.

CERTIFICATIONS

Claude Code in Action | Anthropic | Aug 2026

- Course certificate, Credential ID: t982opugudqo

Claude Code 101 | Anthropic | Aug 2026

- Certificate of completion, Credential ID: icac2urzwu47

VOLUNTEERING

Mentor | Google Developer Groups (GDG) | Jun 2021 - Jun 2024

- Mentor software developers across South and Southeast Asia to grow their Android development skills.

EDUCATION

Bachelor of Science, Computer Science and Engineering

Ahsanullah University of Science and Technology | Dhaka, Bangladesh | 2007 - 2012